

Professional Summary

Artificial Intelligence Engineering student with experience developing software solutions for enterprise environments. I have contributed to full-stack applications, AI-powered conversational systems, and data analysis platforms that support decision-making and optimize operational processes. My professional focus is building products that combine software engineering and artificial intelligence to solve real-world challenges and create value for organizations and users.

Technical Skills

LLMs and agents: conversational agents, RAG, prompt engineering, tool/function calling

AI and ML: computer vision, OCR, multimodal validation, NLP, object classification

Programming languages: Python, JavaScript, TypeScript, SQL, Java, C

Backend: REST APIs, FastAPI, Node.js, Express.js, Spring Boot

Frontend: SAP UI5/Fiori, React, Angular, HTML, CSS

Databases: SAP HANA, PostgreSQL, MySQL, MongoDB

Cloud and tools: Google Cloud Run, SAP BTP, Cloud Foundry, Git, Docker, CI/CD

Projects

Enterprise Technology Recycling Management Platform – AMX Contenido

Node.js, Express, SAP HANA, SAP UI5/Fiori, Google Cloud, JWT

- Developed full-stack modules to manage the deinstallation, collection, recycling, and analysis workflow for technology equipment, integrating roles, statuses, traceability, filters, and operational reports.
- Integrated multimodal AI workflows to extract data from labels and nameplates, compare images against diagrams/catalogs, and retrieve candidates through RAG to validate equipment models.
- Automated bulk upload, digitization, and standardization of 2,000+ PDF reports, consolidating 50,000+ pieces of equipment in about one hour of execution.
- Implemented Word/PDF document generation, Excel exports, and automated calculations for weight, recyclable material costs, and avoided CO₂ emissions.

Inventory Analysis System with Conversational AI Agent – AMX Contenido

FastAPI, Gemini, SQLite, JavaScript, Chart.js, Pandas, OpenPyXL

- Designed an LLM-based multi-agent architecture with separated responsibilities for planning, SQL generation, conversational responses, and interface action execution through structured JSON contracts.
- Developed a conversational dashboard to query inventory in natural language and render responses, tables, charts, KPIs, suggestions, and Excel exports.
- Optimized analysis of 250,000+ records through SQL queries, server-side pagination, operational filters, and on-demand reports.

Individual Jaguar Identification System for Camera Traps – AMX Contenido

Python, PyTorch, OpenCV, YOLO/MegaDetector, SpeciesNet, SAM 2, MegaDescriptor, SIFT, SQLite, Streamlit

- Built a computer vision pipeline to analyze camera-trap videos, detect animal presence, filter candidate species, and generate a catalog of potential individual jaguars.
- Integrated detection, classification, segmentation, and visual embedding models to extract animal crops, compare similarity across appearances, and support individual identification through automatic merging and human review.
- Optimized the processing workflow with early rejection of empty videos, adaptive frame sampling, visual deduplication, GPU batch processing, and checkpoints to resume incomplete runs.

Smart Academic Portal with Conversational Assistant

Node.js, Express, PostgreSQL, React, LLMs, RAG

- Developed an academic platform with modules for students, professors, and administrators, integrating authentication, roles, and REST APIs.
- Implemented a conversational assistant with RAG to answer institutional questions by combining regulations, procedures, and user data.

Multi-Agent Simulation System with Reinforcement Learning

Python, NumPy, Pygame

- Designed a multi-agent simulation environment to model autonomous decision-making, cooperation, competition, and adaptation in dynamic scenarios.
- Implemented Q-Learning, epsilon-greedy strategy, reward shaping, A*, dynamic risk maps, and genetic algorithms to optimize behaviors.

Education

- ESCOM – IPN
B.Eng. in Artificial Intelligence Aug 2023 – Present
- CECyT 9 Juan de Dios Bátiz – IPN
Technical Degree in Programming Aug 2020 – Jul 2023

Certifications

- SQL with Oracle MySQL Server – ONE
- Scientific Computing with Python
- Oracle Next Education F2 T6 Back-end

Languages

- Spanish: Native
- English: Intermediate (B2)